

Measuring Street Walkability Perception in Tel Aviv Using Municipal Open Data

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Code: <https://github.com/dpartuk/smartTraffic>

Abstract

This project investigates whether municipal open data — business registries, complaint logs, demographics, and traffic speeds — can proxy for visual walkability perception, traditionally measured through street-level imagery and human surveys. Using Tel Aviv as a case study, we integrate 8 datasets (370,000+ records, plus 50,947 traffic segments across 24 months), engineer 27 features across 8 categories, and model their relationship to traffic speed as a walkability proxy. Stepwise OLS regression shows that municipal features alone ($\text{Adj-R}^2 = 0.212$) explain more speed variance than traffic patterns alone ($\text{Adj-R}^2 = 0.178$); combined, they reach 0.354. A Random Forest achieves $\text{R}^2 = 0.489$ under 5-fold cross-validation — within the range Li et al. (2022) achieved with deep learning on street imagery (0.41–0.59). Two walkability scores are developed — a PCA composite and a transparent weighted formula — with $r = 0.950$ correlation. Mapping our features to Li et al.'s six VWP categories shows strong alignment in four dimensions and reveals novel insights (e.g., female population growth as a safety proxy) invisible to cameras

1. Introduction and Goals

Walkability - the degree to which a street environment supports and encourages walking - has become a central concern in urban planning, public health, and transportation policy. Walkable neighborhoods are associated with higher physical activity levels, lower carbon emissions, stronger local economies, and improved social cohesion (Speck, 2012; Southworth, 2005).

Measuring walkability, however, remains challenging. Traditional approaches fall into two categories: (1) survey-based methods, where trained auditors or residents rate street segments on standardized scales; and (2) image-based methods, where deep learning models analyze street-view photographs to extract visual features such as vegetation, sidewalk presence, and vehicle density (Li et al., 2022; Dubey et al., 2016). Both approaches are expensive, difficult to scale, and impractical for cities without comprehensive street-level imagery coverage.

This project asks a different question: **can a city's operational data — business registrations, citizen complaint logs, demographic records, socioeconomic indices, and traffic speed measurements — reveal how walkable its streets feel, without any photographs or surveys?**

Our goals are:

1. **Data integration:** build a reproducible pipeline that merges 8 heterogeneous municipal datasets into a unified spatial database, overcoming encoding inconsistencies, coordinate system differences, and Hebrew text normalization challenges.
 2. **Feature engineering:** compute 27 features per traffic segment across 8 thematic categories (traffic, business, street events, complaints, construction, demographics, housing, socioeconomics).
 3. **Modeling:** compare traffic-only, municipal-only, and combined models to test whether municipal data adds explanatory power beyond traffic patterns alone.
 4. **Walkability scoring:** develop two complementary scoring systems — PCA-based and formula-based — that rank every segment and neighborhood in Tel Aviv on a 0–100 walkability scale.
 5. **Validation:** map our municipal features to the six VWP dimensions from Li et al. (2022) and identify alignments, gaps, and novel contributions.
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2. Literature Review

2.1 The Base Paper: Li, Yabuki & Fukuda (2022)

Our project builds directly on Li, Yabuki, and Fukuda (2022), who developed a comprehensive framework for measuring Visual Walkability Perception (VWP) using panoramic street-view images, virtual reality, and deep learning.

Methodology. The authors constructed the VRVWPR dataset: 2,642 panoramic street-view images (5376×2688 pixels) from seven major cities (New York, Los Angeles, London, Paris, Berlin, Tokyo, Osaka). Thirty participants wearing VR headsets performed pairwise comparisons (20,549 total comparisons), scoring each image on six walkability dimensions derived from Alfonzo's hierarchy of walking needs (Alfonzo, 2005). A DenseNet-based deep convolutional neural network (VWPCL model) was trained on these ratings, achieving 85.4% classification accuracy across Low/Medium/High categories.

Feature extraction. DeepLabv3+ semantic segmentation was applied to extract area ratios of 19 physical components from each image: vegetation, road, sidewalk, terrain, building, wall,

fence, sky, car, truck, bus, train, motorcycle, bicycle, pole, traffic light, traffic sign, person, and rider.

Key findings. Stepwise multiple linear regression of these 19 visual features against VWP scores yielded R^2 values of 0.41–0.59 across the six categories. Vegetation emerged as the strongest positive predictor for safety and pleasurability; sidewalk area ratio was the strongest for accessibility; truck and motorcycle presence were consistently negative across all dimensions.

The six VWP categories (from Alfonzo's hierarchy):

1. **Walkability:** overall visual impression of walking support
2. **Feasibility:** incentive factors (land use diversity, facilities)
3. **Accessibility:** visible barriers (dead ends, restricted access)
4. **Safety:** appearance of safety from crime and traffic
5. **Comfort:** street furniture, sidewalk width, amenities
6. **Pleasurability:** appeal, diversity, liveliness

Our project takes the same six-category framework but replaces pixel-level visual features with record-level municipal data features.

2.2 Walkability Measurement: A Brief Survey

The academic and practitioner literature offers several established approaches to walkability scoring:

Walk Score (Front Seat Management, 2007). The most widely adopted commercial walkability index. Walk Score calculates a 0-100 rating based on walking distance to nearby amenities (grocery stores, restaurants, schools, parks), penalizing locations that require long walks. It uses a distance-decay function where amenities within 400 meters (a 5-minute walk) receive full credit, with linearly declining credit up to 1,600 meters. Walk Score correlates well with pedestrian activity but ignores street-level quality, a segment with many nearby amenities connected by a hostile highway interchange scores as highly as one with tree-lined boulevards.

Frank Walkability Index (Frank et al., 2010). A research-standard composite index defined as:

$$\text{Walkability} = (6 \times z_{\text{intersection_density}}) + (z_{\text{net_residential_density}}) + (z_{\text{land_use_mix}}) + (2 \times z_{\text{retail_floor_area_ratio}})$$

Each component is standardized to a z-score, then combined with predetermined weights reflecting their relative importance in predicting walking behavior. The Frank Index has been used extensively in health research to link built-environment features to physical activity and

BMI outcomes. Its transparency and reproducibility make it the standard reference for formula-based walkability indices.

EPA National Walkability Index (NWI). The U.S. Environmental Protection Agency's Smart Location Database ranks census block groups on a 1–20 scale using intersection density, proximity to transit, and employment/household mix. It covers the entire United States but operates at a coarser spatial resolution than Walk Score or the Frank Index.

Place Pulse / Streetscore (MIT Media Lab; Dubey et al., 2016). A crowdsourced perception study where online participants compared pairs of street-view images on six subjective dimensions (safe, lively, boring, wealthy, depressing, beautiful). The resulting dataset trained deep learning models to predict perceived safety and liveliness from images. Place Pulse demonstrated that human perception of streets can be predicted from visual features, but requires extensive crowdsourcing and image coverage.

Pedestrian Environment Review System (PERS; TfL, 2003). A field audit tool used by Transport for London to rate street segments on 29 attributes across four themes: personal safety, crossing quality, footway quality, and wayfinding. PERS produces detailed segment-level assessments but requires trained surveyors and is labor-intensive.

Our contribution. Unlike all of the above, our approach uses *no images, no surveys, no distance-to-amenity calculations, and no field audits*. We extract walkability signals entirely from municipal administrative data that cities already collect for operational purposes — business registries, citizen complaint systems, demographic records, and traffic sensors. This makes the approach scalable to any city that publishes similar open data.

3. Our Project

3.1 Approach

We hypothesize that municipal data contains latent walkability signals: streets with many restaurants, diverse businesses, and few noise complaints feel more walkable than streets with high car ownership and frequent infrastructure disruptions. We use average traffic speed as the dependent variable (proxy for walkability), then extract model-learned feature weights to build a standalone walkability index.

3.2 Traffic Speed as a Walkability Proxy

A central methodological choice in this project is the use of average traffic speed as the dependent variable, a proxy for walkability rather than a direct measure of it. This subsection explains the rationale, the supporting evidence, and the limitations of this choice.

The causal logic. Walkable streets share a set of physical and social characteristics — narrow lanes, mixed-use frontages, high pedestrian volumes, street-level commerce, traffic calming infrastructure, that collectively create friction for motor vehicles. The result is chronically slower traffic. This is not congestion in the rush-hour sense (temporary demand exceeding capacity), but a structural property of the street environment: streets designed for people naturally resist fast driving. Average traffic speed, measured across all weekday hours (6 AM to 8 PM), captures this structural slowness while smoothing out temporary congestion spikes.

Literature support. Several lines of evidence support this proxy:

- *Shared space design.* The "shared space" movement in urban design (Hamilton-Baillie, 2008) demonstrates that removing traffic signs, curbs, and lane markings in pedestrian-priority areas naturally reduces vehicle speeds to 15–20 kph — precisely because drivers must negotiate space with pedestrians. The design principle confirms the relationship: more walkable design produces slower traffic. Notably, Tel Aviv's most walkable corridor (Rothschild Boulevard area) averages 11.6 kph — well below even shared-space design targets — while the least walkable segments (Neve Dan, Ramat HaChayal) average 17–18 kph.
- *Eyes on the street.* Jane Jacobs (1961) observed that lively, mixed-use streets with active ground-floor commerce generate continuous pedestrian presence, which creates friction that slows vehicles. Our `restaurant_count` and `total_businesses` features capture exactly this mechanism — and their strong negative correlation with traffic speed confirms it empirically.
- *Speed and pedestrian safety.* Research on pedestrian injury risk (Tefft, 2013) shows that the probability of severe pedestrian injury rises sharply above 30 kph. Streets where average speeds remain low (the Tel Aviv central segments average 11–15 kph) are inherently more compatible with walking. Speed is not merely correlated with walkability — it is a *component* of it.

Why not use a direct walkability measure? Four alternatives exist in the literature, and none are available for Tel Aviv at the required spatial resolution:

1. *Human perception surveys* (as in Li et al., 2022) would provide the most direct ground truth, but no such survey exists for Tel Aviv at the segment level.
2. *Street-view imagery* could be analyzed with deep learning, but Google Street View coverage in Tel Aviv is incomplete and access to panoramic imagery at scale is proprietary and costly.
3. *Walk Score* measures proximity to amenities via network distance — it captures *access* but not *street-level experience*. A segment next to a highway on-ramp may score well on Walk Score because shops are nearby, yet feel hostile to walk.
4. *Field audits* (e.g., PERS) require trained surveyors and are impractical for 2,070 segments.

Traffic speed is the only continuously measured, segment-level, city-wide variable available in Tel Aviv's open data. It is recorded for every segment, every hour, every month — giving us 50,947 segment-months of coverage.

Guarding against false signals. The main risk of this proxy is that some streets are slow for reasons unrelated to walkability — highway merge zones during rush hour, construction bottlenecks, or signal-heavy arterials. We mitigate this in three ways:

1. *Temporal averaging.* We use the mean speed across all weekday hours (6 AM–8 PM), not peak-hour speed. Chronic slowness across the full day is a walkability signal; temporary morning congestion is not.
2. *Multi-feature modeling.* Speed is the dependent variable, not the sole walkability measure. The model learns which municipal features (restaurants, complaints, demographics) predict slow streets — if a street is slow only because of a construction detour, it will lack the municipal feature signature of a genuinely walkable street.
3. *PCA and weighted index.* Our walkability scoring systems use 24 features (PCA) or 10 features (weighted index), not speed alone. Speed is one input among many, and its influence is modulated by the other features.

Empirical validation. The neighborhoods ranked highest by our model — Lev Tel Aviv (Rothschild Boulevard area, score 93.6), Old North (73.1), Florentin (45.9) — are widely recognized as Tel Aviv's most walkable areas by residents, tourists, and urban planning assessments. The lowest-ranked neighborhoods — Neve Dan (2.9), Rabivim (4.9), Ramat HaChayal (9.0) — are car-dependent peripheral areas with limited pedestrian infrastructure. This face-validity alignment suggests that, despite its indirectness, traffic speed combined with municipal features produces rankings consistent with lived experience.

3.3 Datasets

All data are freely available from the Tel Aviv Open Data Portal (opendata.tel-aviv.gov.il) and the Tel Aviv ArcGIS server. The datasets span traffic, commercial activity, civic engagement, demographics, housing, and socioeconomics.

Dataset	Records	Source	Format	Key Fields
Traffic speeds	50,947 segments × 24 months	ArcGIS REST API	GeoJSON	Hourly avg speed (6 AM–8 PM), segment geometry
Business registry	~28,000	Azure Blob	CSV (UTF-16LE)	Category, neighborhood,

Dataset	Records	Source	Format	Key Fields
				licensing status, area
106 Hotline complaints	~180,000	Azure Blob	CSV (UTF-16LE)	Category (3-level), neighborhood, year/month
Closed streets	~8,500	Azure Blob	CSV (UTF-16LE)	Closure type (event/rail/infra), dates, street
Construction permits	~15,000	Azure Blob	CSV (Win-1255)	Year, district, type (start/completion), dwellings
Population by age/gender	~85,000	Municipal server	CSV (Win-1255)	Age group, gender, neighborhood, year
Dwelling units	~4,500	Azure Blob	CSV (Win-1255)	Residential/commercial, neighborhood
Socioeconomic index	71 neighborhoods	Municipal server	Excel	18 SES indicators including car ownership, income

Technical challenges included 3 encodings, 4 delimiters, ArcGIS pagination (2,000-record limit), inconsistent Hebrew spelling across datasets, and sub-district-to-neighborhood mapping. All data is loaded into a GeoPackage (16 layers, 243 MB).

Spatial database. All datasets are loaded into a GeoPackage (smarttraffic.gpkg) containing 16 layers, 3 spatial (traffic segments for all months, latest month, and closed streets) and 13 attribute tables. Total size: 243 MB.

3.4 Feature Engineering

The unit of analysis is a directed traffic segment (~100 m). Segments are spatially joined to 71 neighborhoods via centroid point-in-polygon matching (EPSG:2039 → 4326, 99.6% match rate). We compute 27 features:

Category	Features	Sign	Join
Traffic (5)	avg_speed, peak_speed_drop, speed_variance, dir_speed_diff, evening_speed_reco very	-, -, -, -, +	Direct
Business (5)	restaurant_count, total_businesses, biz_diversity_index (Shannon entropy), licensed_biz_ratio, evening_biz_count	all +	Neighborhood
Street Events (5)	event_frequency, event_diversity, light_rail_closures, parade_route_count, infra_closure_count	+, +, -, +, -	Street code
Complaints (2)	noise_complaints, infra_complaints	both -	Neighborhood
Construction (2)	construction_starts, construction_completi on_rate	-, +	Sub-district map
Demographics (3)	female_pop_growth_ yoy, young_adult_share, elderly_share	+, +, -	Hebrew name
Housing (2)	residential_density, commercial_to_resid ential_ratio	+, neutral	Neighborhood

Category	Features	Sign	Join
Socioeconomic (3)	ses_cluster, cars_per_100_residents, avg_monthly_income	+, -, neutral	Neighborhood

Coverage: 82–100% across all features. Final dataset: 2,070 segments with complete traffic data; 1,278–1,408 with full municipal coverage.

Key libraries: GeoPandas (spatial operations), statsmodels (OLS regression, VIF), scikit-learn (ML models, PCA, permutation importance), Folium (interactive maps), matplotlib/seaborn (visualization).

3.6 Results

3.6.1 Multicollinearity Diagnostics

Exploratory analysis begins with feature distributions (Figure 1) and a correlation heatmap (Figure 2) across all 27 features. A Variance Inflation Factor (VIF) check identified 9 features with $VIF > 10$, indicating severe multicollinearity. The most extreme case was evening_biz_count ($VIF = 1,498$), which correlates at $r = 0.999$ with restaurant_count — both measure the same underlying signal of neighborhood food/entertainment vitality. Stepwise regression (forward/backward with $p_{\text{enter}} = 0.05$, $p_{\text{remove}} = 0.10$) was used to handle multicollinearity by selecting only features that add statistically significant explanatory power.

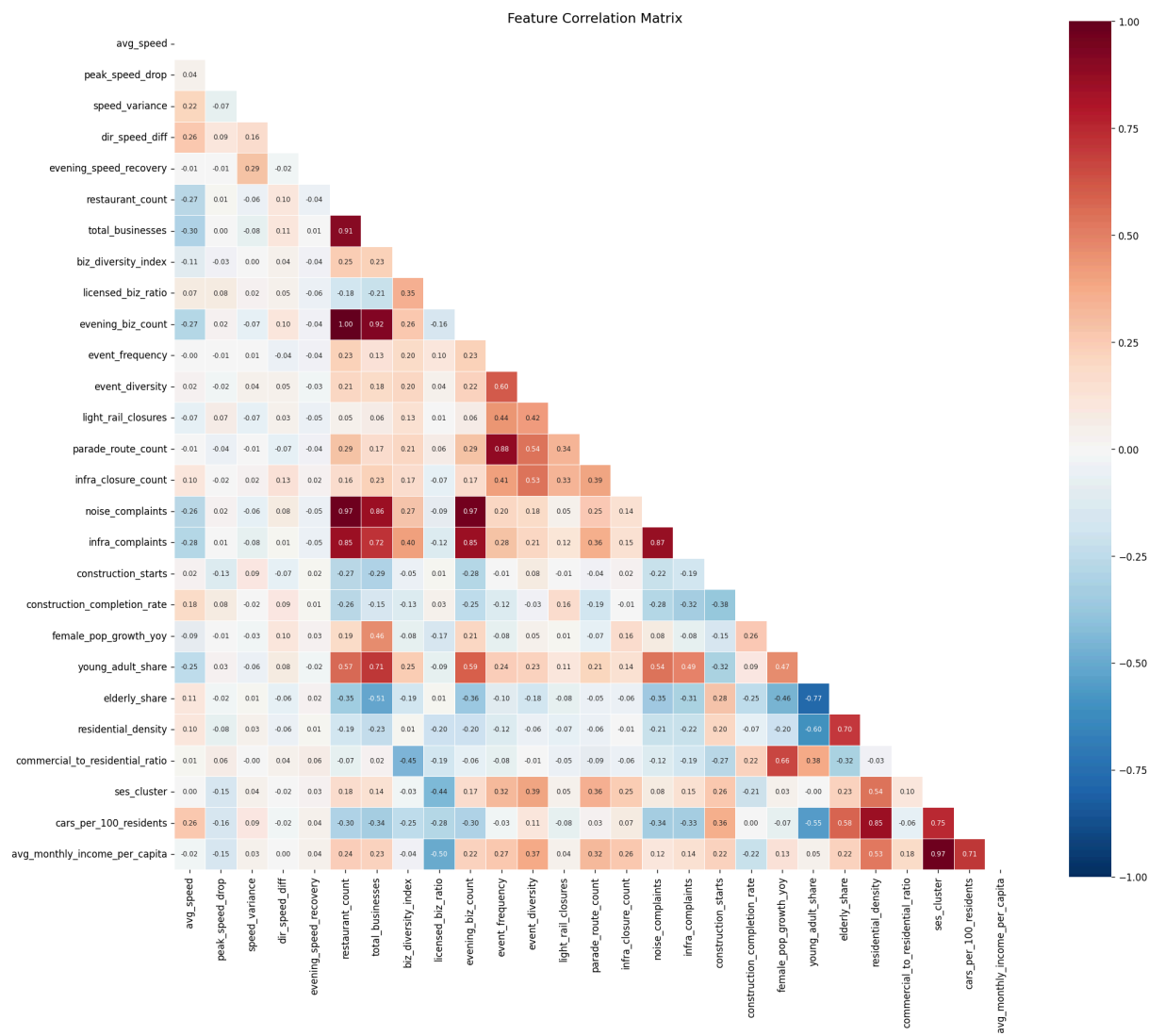
3.6.2 Linear Regression Models

Three stepwise OLS regression models were fitted to predict average traffic speed:

Model	Predictors	Features	Adj-R ²	Key Finding
A: Traffic only	4	speed_variance, peak_speed_drop, dir_speed_diff, evening_speed_recovery	0.178	Traffic patterns alone explain 17.8% of speed variance
B: Municipal only	12	restaurant_count, ses_cluster, cars_per_100,	0.212	Municipal data alone explains 21.2% — more than traffic

Model	Predictors	Features	Adj-R ²	Key Finding
		noise_complaint s, ...		
C: Combined	17	Both traffic and municipal features	0.354	Combined model explains 35.4%

The central finding: Model B (municipal only) outperforms Model A (traffic only), demonstrating that municipal administrative data captures walkability-relevant information that traffic sensors alone do not. The combined Model C achieves an Adj-R² improvement of +0.142 over municipal-only, confirming that traffic and municipal features provide complementary signals. Regression coefficients for all three models are shown in Figure 3.



3.6.3 Machine Learning Models

The stepwise OLS models above identify which feature groups matter and compare their explanatory power. To evaluate how well these features generalize to unseen data, we trained three machine learning models using 5-fold cross-validation on the combined 27-feature set:

Model	CV R ² (mean ± std)	CV MAE	Notes
Linear Regression	0.330 ± 0.029	2.3 kph	Stable baseline
Neural Network	0.437 ± 0.118	2.0 kph	+32% over LR, but high variance
Random Forest	0.489 ± 0.022	1.9 kph	Best accuracy, most stable

The Random Forest achieves the highest R² (0.489) with the lowest standard deviation (0.022), making it both the most accurate and most reliable model. For comparison, Li et al. (2022) achieved R² = 0.41-0.59 using street-level imagery and deep learning - our result falls within that range using only tabular open data.

The Neural Network's high variance (±0.118) reflects insufficient training data for deep learning (only ~1,400 complete samples). The Random Forest's stability and accuracy without hyperparameter tuning make it the most practical choice for municipal deployment.

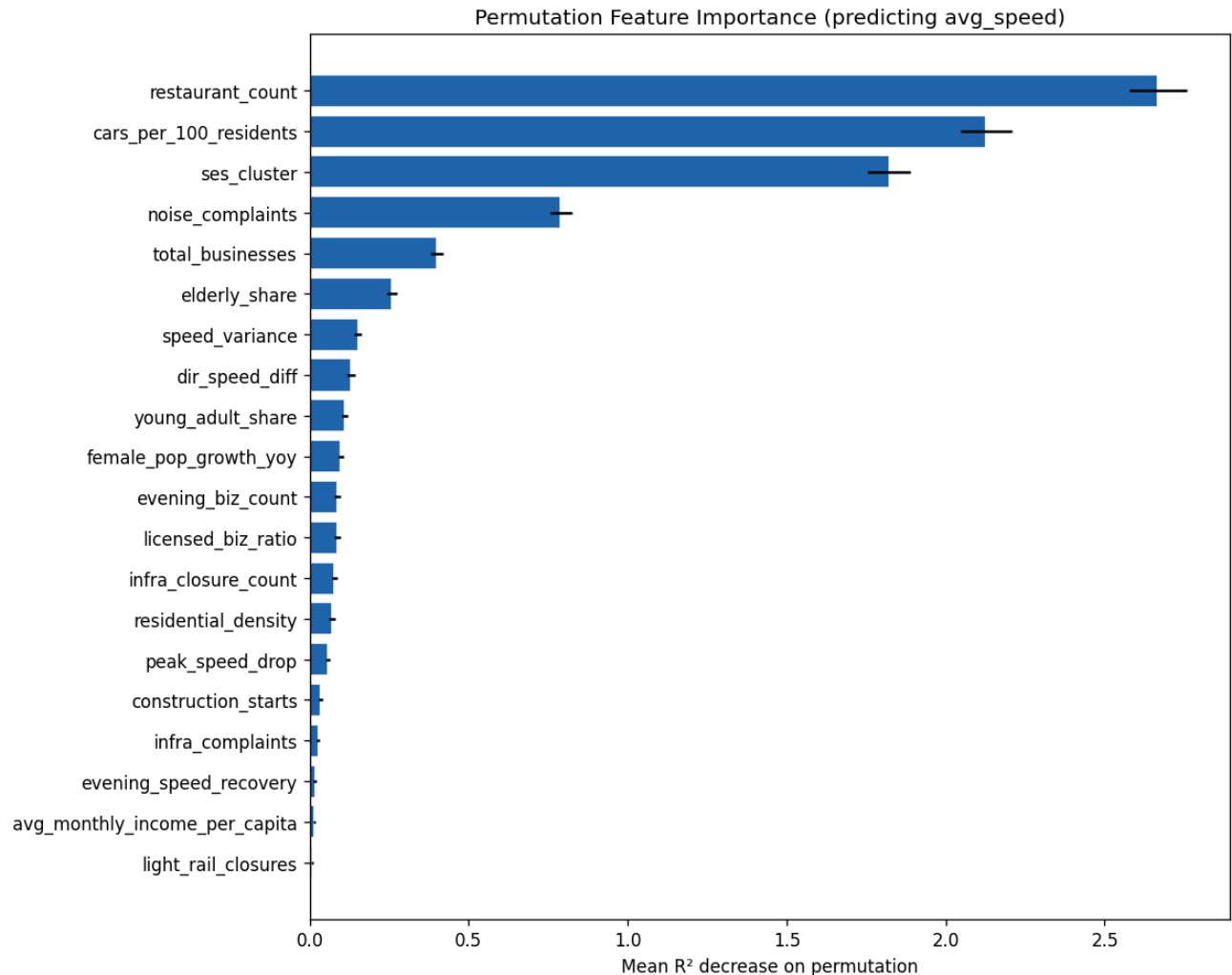
3.6.4 Feature Importance

Permutation importance analysis reveals the top drivers of walkability:

Rank	Feature	Importance (R ² decrease)	Interpretation
1	restaurant_count	2.668	Street vitality — restaurants attract foot traffic
2	cars_per_100_residents	2.127	Car dependency — more cars = less walking
3	ses_cluster	1.822	Socioeconomic status — wealthier

Rank	Feature	Importance (R ² decrease)	Interpretation
			areas have better infrastructure
4	noise_complaints	0.790	Pedestrian discomfort — noise deters walking
5	total_businesses	0.400	Mixed use — diverse commerce attracts pedestrians
6	elderly_share	0.259	Age demographics — higher elderly share = less active streets
7	speed_variance	0.152	Traffic instability — unpredictable speeds feel unsafe
8	dir_speed_diff	0.130	Directional asymmetry — indicates bottlenecks
9	young_adult_share	0.111	Young adults prefer and seek walkable areas
10	female_pop_growth	0.098	Women move to neighborhoods that feel safe

The dominance of municipal features (8 of top 10) over traffic features (2 of 10) reinforces the finding that municipal data is a richer source of walkability information than traffic sensors alone (Figure 7).



3.6.5 From Speed Model to Walkability Score

The regression and machine learning models described above predict *traffic speed*, not walkability. This subsection explains the logical bridge between the two.

The models serve two purposes, and the second is more important than the first:

1. **Primary purpose: evidence.** The models prove that municipal features carry a statistically significant signal about street character. Model B's Adj-R² = 0.212 exceeding Model A's 0.178 is the core finding - it demonstrates that business registries, complaint logs, and demographics explain more about a street's traffic behavior than the traffic data's own patterns.
2. **Secondary purpose: feature weighting.** As a byproduct of training, the models reveal *which features matter most* and *by how much*. Permutation importance quantifies this: shuffling `restaurant_count` destroys 2.668 units of R², while shuffling `dir_speed_diff`

destroys only 0.130. These importance values are not speed predictions - they are measures of each feature's *information content* about street character.

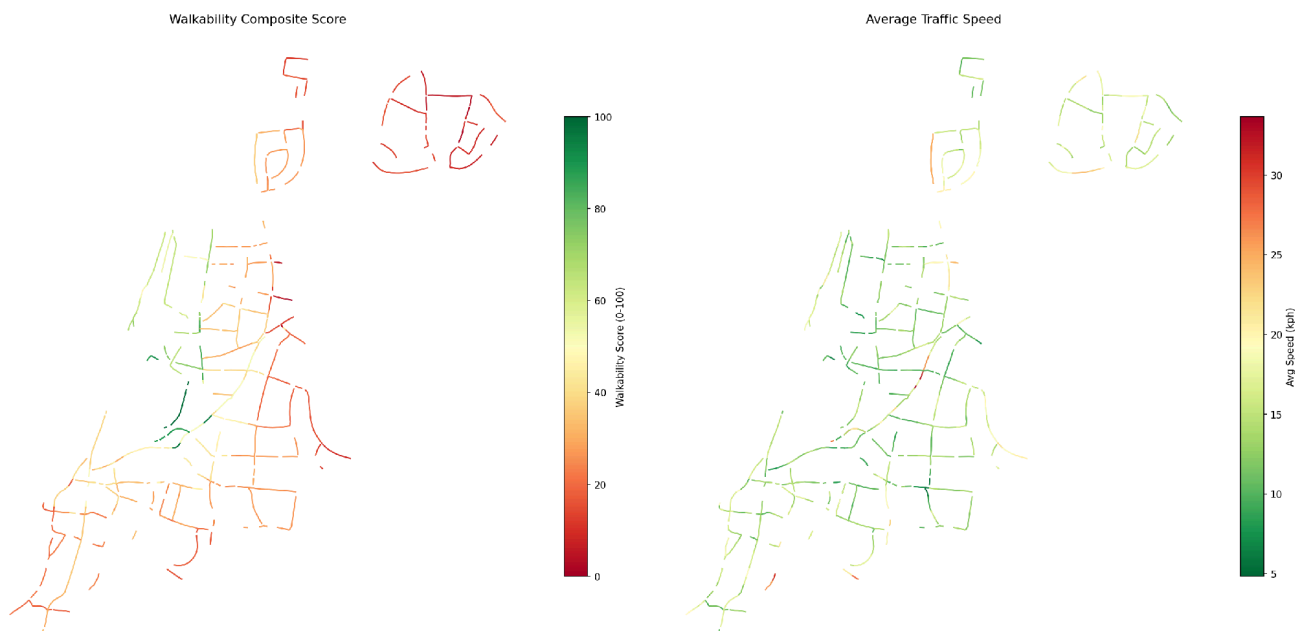
The walkability formula exploits purpose (2). We extract the importance values from the trained model and repurpose them as weights in a standalone index. The model itself - its coefficients, its speed predictions - is discarded. Only its learned knowledge about *what matters* survives into the formula.

The logic chain is:

Train model (features → speed) → **Extract importance** (which features carry signal) → **Assign signs** (domain knowledge: restaurants are positive, noise is negative) → **Build formula** (importance-weighted z-scores) → **Walkability score** (0–100)

This is analogous to how the Frank Walkability Index was constructed: Frank et al. (2010) used regression analysis of walking behavior to determine that intersection density matters more than residential density, then assigned intersection density a weight of 6× in their formula. We follow the same logic - data-driven weight discovery, domain-driven sign assignment - but with municipal features instead of built-environment GIS variables.

Tel Aviv Walkability Analysis



3.6.6 Walkability Scoring — PCA Approach (Approach 1)

Principal Component Analysis was applied to the 24 features with known walkability direction (positive = more walkable, after sign-flipping). PC1 explains 26% of total variance and captures the primary walkability gradient (Figure 4, scree plot and loadings). PC1 scores are normalized to a 0–100 scale to produce the walkability index.

The resulting walkability map of Tel Aviv (Figure 5) reveals a clear geographic pattern: central, mixed-use neighborhoods score high; peripheral, car-dependent neighborhoods score low. Table 1 presents the top and bottom neighborhoods.

Table 1. Neighborhood walkability rankings (PCA score, 0–100 scale).

Rank	Neighborhood	Score	Avg Speed (kph)	Key Characteristics
1	Lev Tel Aviv	93.6	15.5	Rothschild area — highest restaurant density, high SES
2	Old North (south)	73.1	11.4	Dense mixed-use, high young adult share
3	Old North (north)	62.1	12.6	Residential-commercial mix, frequent events
4	Montefiore	48.0	13.2	Historic mixed-use, moderate business density
5	Florentin	45.9	12.1	Arts/nightlife district, high evening activity
...				

Rank	Neighborhood	Score	Avg Speed (kph)	Key Characteristics
37	Ramat HaChayal	9.0	15.0	Tech-office district, few restaurants, high car ownership
38	Rabivim	4.9	15.9	Residential, few businesses, few events
39	Neve Dan	2.9	17.9	Peripheral, car-dependent, lowest commercial activity

3.6.7 Walkability Scoring — Weighted Formula Approach (Approach 2)

While PCA is statistically rigorous, its interpretation requires understanding of eigenvalues and loadings. To provide a more transparent and actionable alternative, we developed a weighted z-score index following the structure of the Frank Walkability Index (Section 2.2). Like the Frank Index, we standardize each feature to a z-score and combine them with empirically derived weights. Our adapted formula:

$$\text{Walkability} = \sum (w_i \times \text{sign}_i \times z_score_i), \text{ normalized to } 0\text{--}100$$

Where weights are derived from permutation importance of the linear regression model (Section 3.6.5), signs indicate the expected direction of each feature's effect on walkability, and z-scores standardize features to zero mean and unit variance.

Feature	Sign	Weight	Interpretation
restaurant_count	+	0.312	More restaurants = more street vitality
cars_per_100_residents	–	0.249	Car dependency reduces walkability
ses_cluster	+	0.213	Higher SES = better pedestrian infrastructure

Feature	Sign	Weight	Interpretation
noise_complaints	–	0.092	Noise = pedestrian discomfort
total_businesses	+	0.047	Mixed-use streets attract walking
elderly_share	–	0.030	Higher elderly share = less active streets
speed_variance	–	0.018	Unstable traffic = less safe for pedestrians
dir_speed_diff	–	0.015	Asymmetric traffic flow = bottleneck
young_adult_share	+	0.013	Young adults prefer walkable areas
female_pop_growth	+	0.012	Women move to safe neighborhoods

Validation. The weighted index was computed for 1,408 segments. Its Pearson correlation with the PCA-based score is $r = 0.950$, confirming that both approaches produce nearly identical neighborhood rankings. The top and bottom neighborhoods match between the two methods (Figure 13 shows the signed weight distribution). This high agreement validates the weighted formula as a simpler, more interpretable alternative to PCA — planners can point to specific features and their weights to explain why a neighborhood scores the way it does.

3.7 Comparison with Li et al. (2022)

We map our 27 municipal features to the six VWP categories from Li et al. (2022) and assess alignment:

VWP Category	Li et al. Visual Features	Our Municipal Proxy	Agreement
Walkability	Vegetation, sidewalk, person, bicycle	Business density, restaurant count, residential density	Both identify street vitality as key; our business metrics proxy for person/bicycle presence

VWP Category	Li et al. Visual Features	Our Municipal Proxy	Agreement
Feasibility	Road, traffic light, sign, fence (–)	Business diversity index, commercial-to-residential ratio	Mixed-use diversity aligns with land-use reading; infra_closure proxies for fence/barrier
Accessibility	Sidewalk, fence (–), wall (–)	Infrastructure complaints, construction disruption, light rail closures	Both identify physical barriers; complaints capture sidewalk quality without pixel counting
Safety	Vegetation, truck (–), motorcycle (–)	Female pop growth, evening businesses, noise complaints	Novel: female population growth as safety signal has no visual equivalent
Comfort	Vegetation, car (–), truck (–), sky	Noise complaints, construction, cars per 100 residents	Strong alignment: car density mirrors car/truck pixel ratio
Pleasurability	Vegetation, terrain, person, bicycle	Restaurant density, event frequency, young adult share	Event/restaurant features capture liveliness from person/bicycle pixels

Key insight. Four of six categories (Walkability, Accessibility, Comfort, Pleasurability) show strong alignment between visual and municipal features — the municipal data captures the same walkability dimensions that Li et al. identified through pixel-level image analysis, despite using entirely different input data.

The Safety category reveals a particularly novel contribution. Li et al. found that vegetation and the absence of trucks/motorcycles predict perceived safety. Our municipal equivalent — female population growth — operates through a completely different mechanism: it captures *revealed preference* rather than visual appearance. Women disproportionately avoid neighborhoods perceived as unsafe when choosing where to live; year-over-year female population growth therefore serves as an aggregated safety signal derived from thousands of individual residential decisions. This is a walkability dimension that no camera can see. Similarly, evening business count serves as a proxy for Jane Jacobs' (1961) "eyes on the street" concept — neighborhoods with active restaurants and shops after dark provide natural surveillance that enhances

perceived safety, a signal that street-view images (typically captured during daytime) would miss entirely.

4. Discussion

4.1 Municipal Data as a Walkability Proxy

Our results demonstrate that municipal administrative data captures walkability dimensions that complement and sometimes exceed what traffic sensors or street-level imagery can measure. The fact that municipal-only features ($\text{Adj-R}^2 = 0.212$) outperform traffic-only features ($\text{Adj-R}^2 = 0.178$) in predicting average traffic speed is notable, because traffic speed is a *traffic* variable — yet the municipal context in which that traffic occurs is more predictive than the traffic patterns themselves.

4.2 Practical Implications

The weighted walkability formula provides an actionable tool for urban planners. Unlike PCA (which requires statistical expertise to interpret) or Walk Score (which is proprietary), our formula is transparent: each feature has a named weight and a clear direction. A planner can immediately see that `restaurant_count` (weight 0.312) dominates the index, suggesting that zoning policies encouraging mixed-use ground floors would have the largest impact on perceived walkability. Similarly, the strong negative weight on `cars_per_100_residents` (−0.249) directly supports traffic calming and parking reduction policies.

4.3 Limitations

1. **No ground truth.** We use average traffic speed as a walkability proxy rather than direct human perception ratings. Section 3.2 details the rationale, supporting evidence, and mitigations for this choice, but the fundamental limitation remains: speed is an indirect measure, and validation against a human perception survey would strengthen the findings.
2. **No visual features.** Street-level imagery was not acquired. Adding vegetation cover, sidewalk width, and building facade data would likely improve model accuracy.
3. **Spatial granularity.** Municipal data (businesses, complaints, demographics) is available at the neighborhood level, so all segments within a neighborhood receive identical municipal feature values. Finer-grained data would improve street-level accuracy.
4. **Temporal coverage.** Construction data extends only through 2022; demographic data through 2022. More recent data would improve currency.
5. **Spatial autocorrelation.** A Durbin-Watson statistic of 1.16 suggests positive spatial autocorrelation — nearby segments have correlated residuals, which may inflate statistical significance.

5. Conclusion

This project demonstrates that municipal open data can serve as a viable proxy for walkability perception. Using 8 datasets, 27 features, and a multi-model analysis pipeline, we achieve $R^2 = 0.489$ (Random Forest) — comparable to the $R^2 = 0.41\text{--}0.59$ that Li et al. (2022) achieved with deep learning on street-view imagery. Our weighted walkability index provides a transparent, reproducible formula that any city publishing business registries, complaint logs, and demographic data can adapt. The approach requires no cameras, no surveys, and no proprietary tools — only publicly available administrative data and open-source Python code.

Future work. Three directions would strengthen and extend these results:

1. *Ground truth validation.* Conducting a human perception survey — even a small one covering 10–15 neighborhoods — would provide direct validation of the walkability scores against subjective experience.
2. *Visual feature integration.* Adding green space and tree canopy data (available from satellite imagery or municipal GIS) would address the most significant gap relative to Li et al.'s visual features, where vegetation was the strongest predictor.
3. *Temporal analysis.* The current model produces a static snapshot. Tracking walkability scores over time — particularly in neighborhoods where new restaurants open, light rail construction completes, or car ownership shifts — would test whether the index captures dynamic changes in street character.

Reproducibility. The entire pipeline is implemented in open-source Python and runs on freely available data. Any city that publishes business registries, citizen complaint logs, demographic records, and traffic speed measurements can adapt the methodology by substituting its own datasets and adjusting the feature engineering joins to match local data structures.

References

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